INCH-POUND

MIL-T-21038/11F
9 March 1990
SUPERSEDING
MIL-T-21038/11E(USAF)
24 July 1987

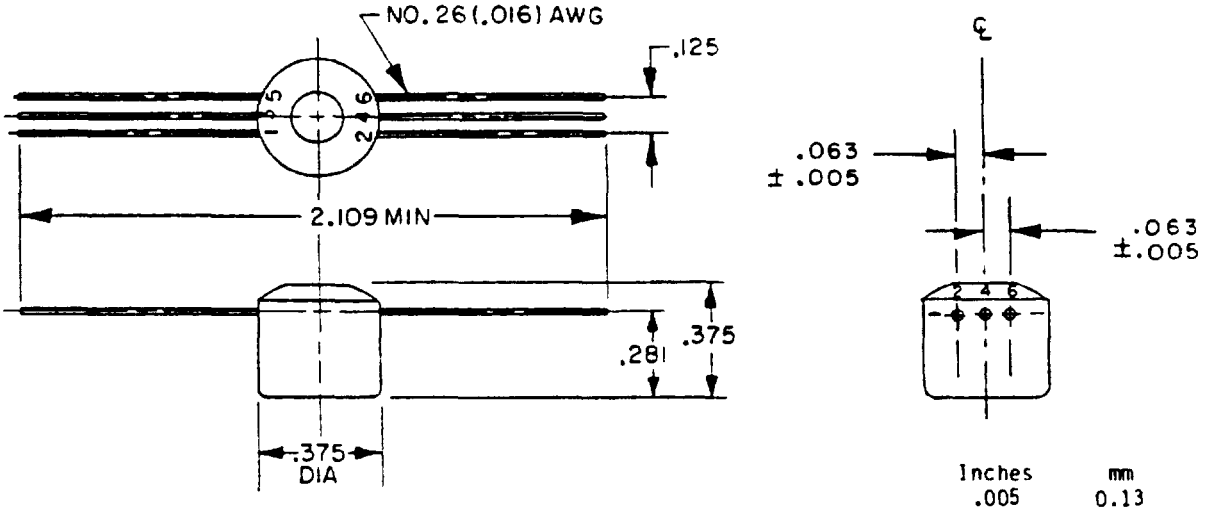
MILITARY SPECIFICATION SHEET

TRANSFORMERS, PULSE, LOW POWER,

TYPE TP7SX4410AZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-T-21038.



Inches	mm
.005	0.13
.016	0 41
.063	1.60
.125	3.18
.281	7.14
.375	9.52
2.109	53 57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only
3. Unless otherwise specified, tolerances are ±.032 (0 81 mm)

FIGURE 1. Dimensions and configuration.

(F) denotes changes

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REQUIREMENTS.

Dimensions and configuration See figure 1.

Case: Encapsulated.

Terminals Solid wire leads, type D in accordance with MIL-STD-1276, tinned

(F) Weight. 2.5 grams maximum.

Turns ratio 4 4 1.

Terminal strength. Method 211, MIL-STD-202, test condition A, pull 2.5 pounds.

Dielectric withstanding voltage

Atmospheric pressure: Method 301, MIL-STD-202, 100 volts rms for 60 seconds between windings and ground.

Barometric pressure (reduced): Method 301, MIL-STD-202 and method 105, test condition B, MIL-STD-202, 100 volts rms between windings and ground.

Insulation resistance Method 302, test condition A, MIL-STD-202, not less than 10,000 megohms.

Electrical characteristics See table 1.

Test circuit See figure 2.

Life (expectancy "X") Applicable

Temperature rise: Not applicable.

Vibration high frequency: Method 204, test condition B, MIL-STD-202.

Shock: Not applicable.

Quality assurance provisions

Extent of qualification Qualification testing and approval to M21038/11-001 shall be sufficient to grant qualification approval to M21038/11-001 through M21038/11-009.

(F) Part or Identifying Number (PIN) M21038/11- (dash numbers from table I). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification.

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TABLE I. Electrical characteristics including waveform parameters

Dash no	DCR ±25% (ohms)			Blocking oscillator pulse					Coupling circuit characteristics							Frequency response 2/ ±2 dB
				Pulse width (µs)	Rise time (µs)	Over-shoot %	Droop %	Back swing %	Pulse width (µs)	Volts out	Rise time (µs)	Over-shoot %	Droop %	Back swing %	Impedance 1/ out, in (ohms)	
	1-2	3-4	5-6													
001	42 0	52 0	15 0	10	14	0	15	28	10	6 5	.4	0	15	30	200	1 1 kHz - 400 kHz
002	15 0	18 0	5 4	5	097	0	23	28	5	7 9	2	0	18	28	200	3 6 kHz - 840 kHz
003	10 0	12.0	3 6	3	087	0	18	30	3	6 8	17	0	10	20	100	3 9 kHz - 950 kHz
004	6 2	7 3	2 2	2	066	0	15	25	2	6 6	14	0	10	20	100	8 5 kHz - 1 G75 MHz
005	3 7	4 4	1 3	1	.033	0	12	25	1	7	078	0	15	23	100	12 kHz - 2 3 MHz
006	1.9	2 3	7	5	027	2	10	20	5	7 5	.045	0	20	25	100	22 kHz - 3 7 MHz
007	8	1 0	3	2	022	0	0	18	2	7 4	034	0	5	12	100	60 kHz - 14 5 MHz
008	4	5	2	1	016	0	0	30	1	8.2	021	0	0	15	50	95 kHz - 17 MHz
009	.12	14	.05	05	016	0	0	30	.05	9.3	012	0	0	20	50	550 kHz - 43 MHz

1/ See figure 2.
2/ Per coupling c rcuit (Z), in/out, 1 V input.

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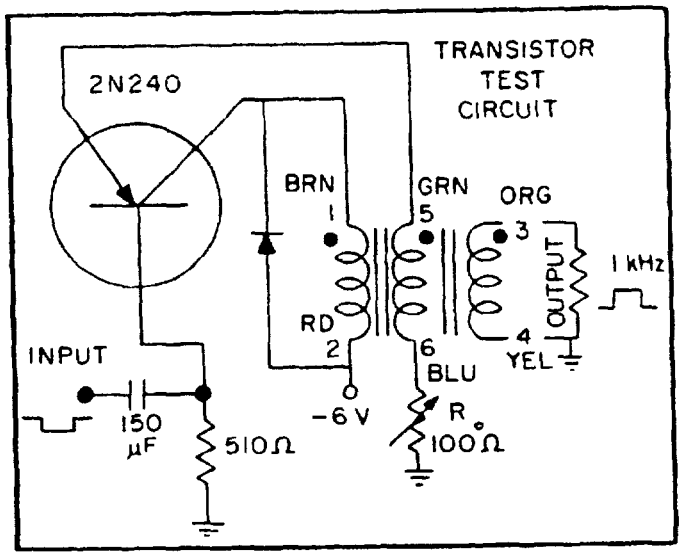


FIGURE 2. Test circuit.

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 17
DLA - ES

User activities
Army - ME
Navy - MC
Air Force - 19

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5950-0742)

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